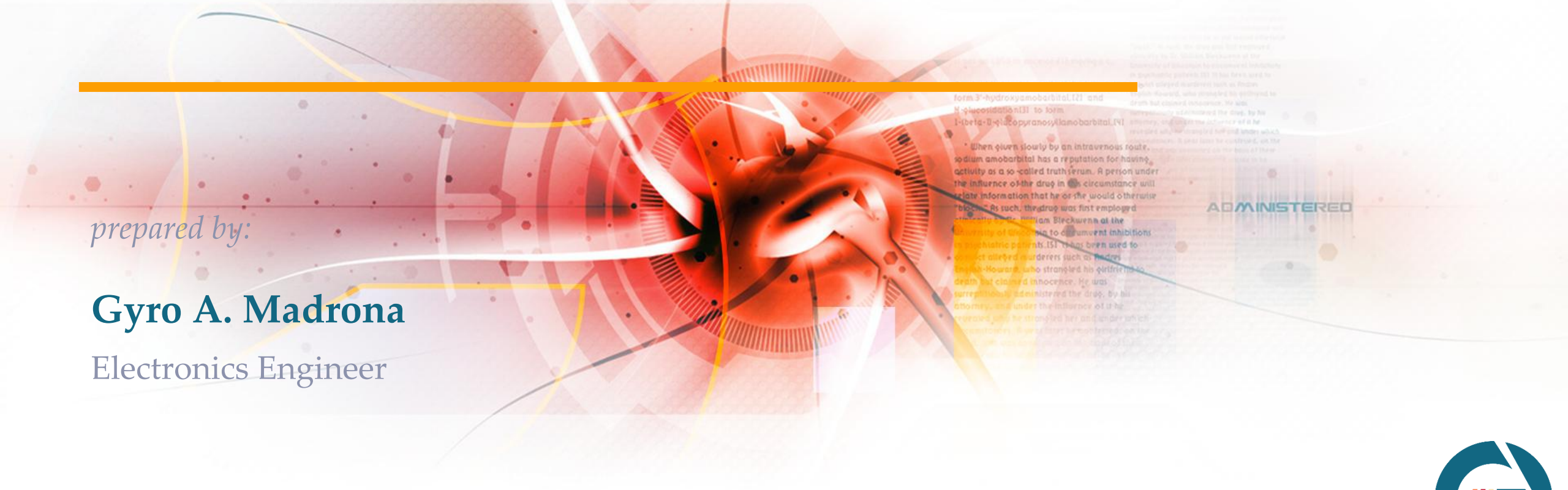
THE CONE

SOLID MENSURATION



prepared by:

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Electronics Engineer

GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

40%

WEEKLY

60%

EXAM

0

ABSENT

3

45-min LATE

5 max

15-min LATE

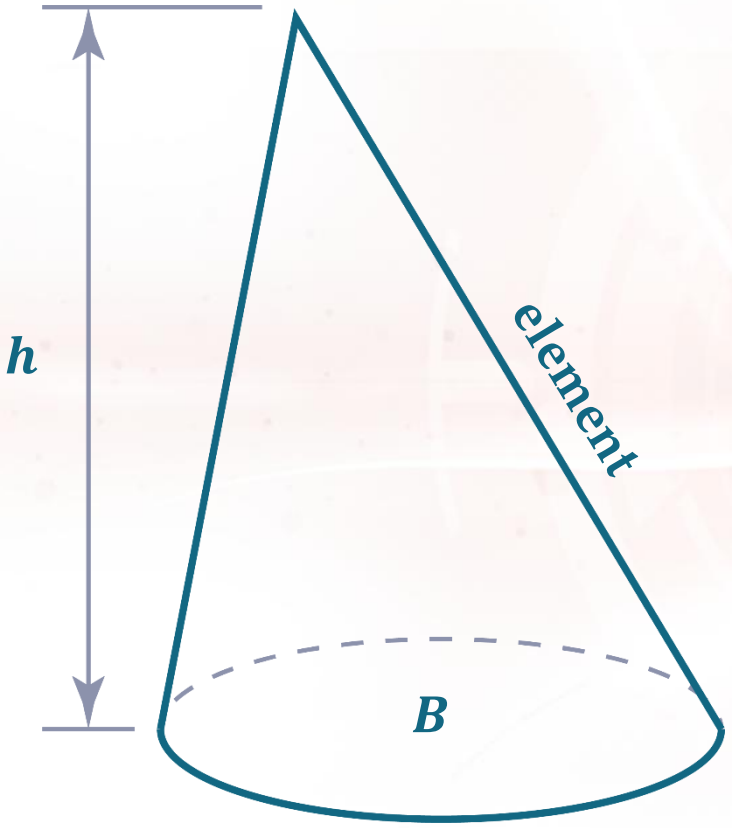
TOPIC OUTLINE

Properties of a Cone

- Lateral Surface Area
- Total Surface Area
- Volume

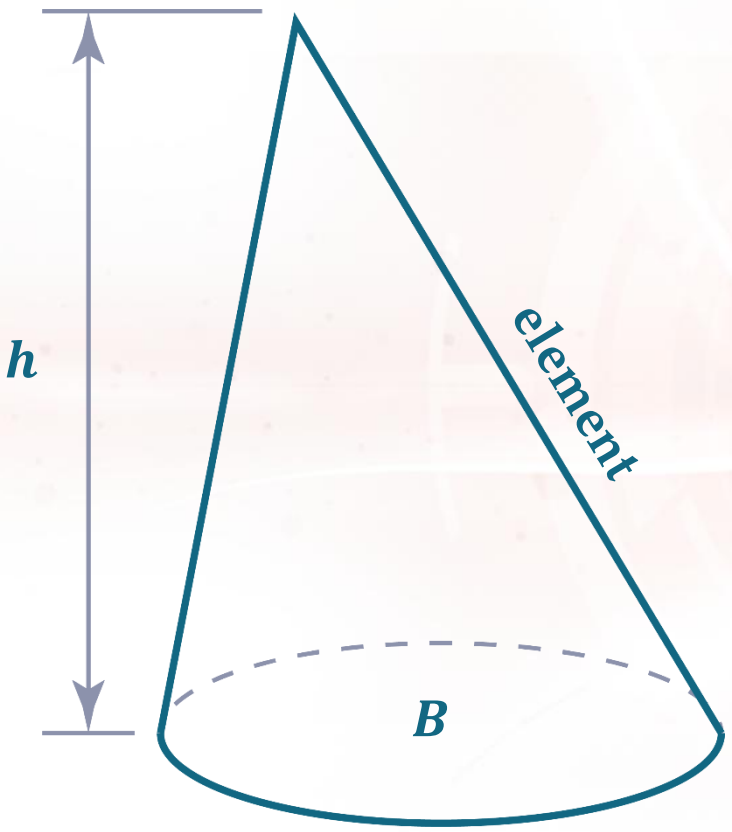
PROPERTIES OF A CONE

DEFINITION



A **cone** is the solid bounded by a conical surface (**lateral surface**) whose directrix is a **closed curve**, and a plane (**base**) which cuts all the elements.

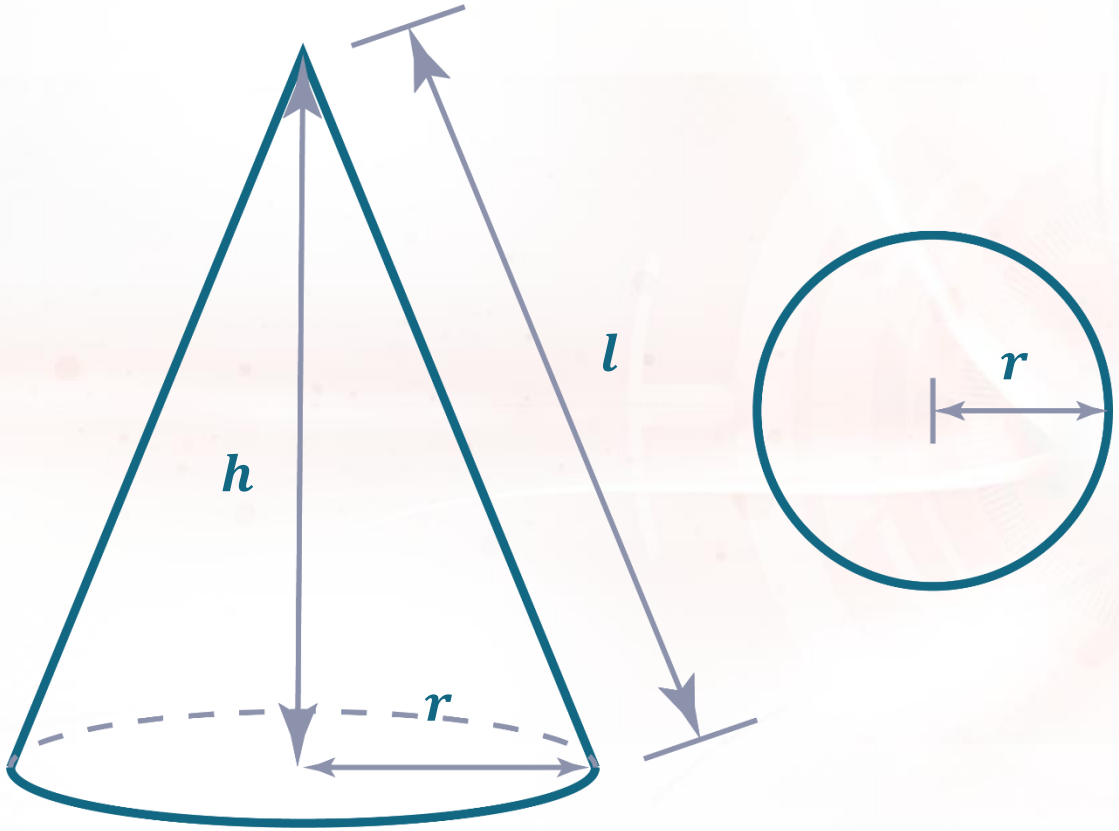
PROPERTIES



1. The **altitude, *h*** of a cone is the perpendicular distance from the vertex to the plane of the base.
2. Every section of a cone made by a plane passing through its vertex and containing two points of the base is a **triangle**.
3. A right section of a cone is a section perpendicular to its axis and cutting all the ***elements***.

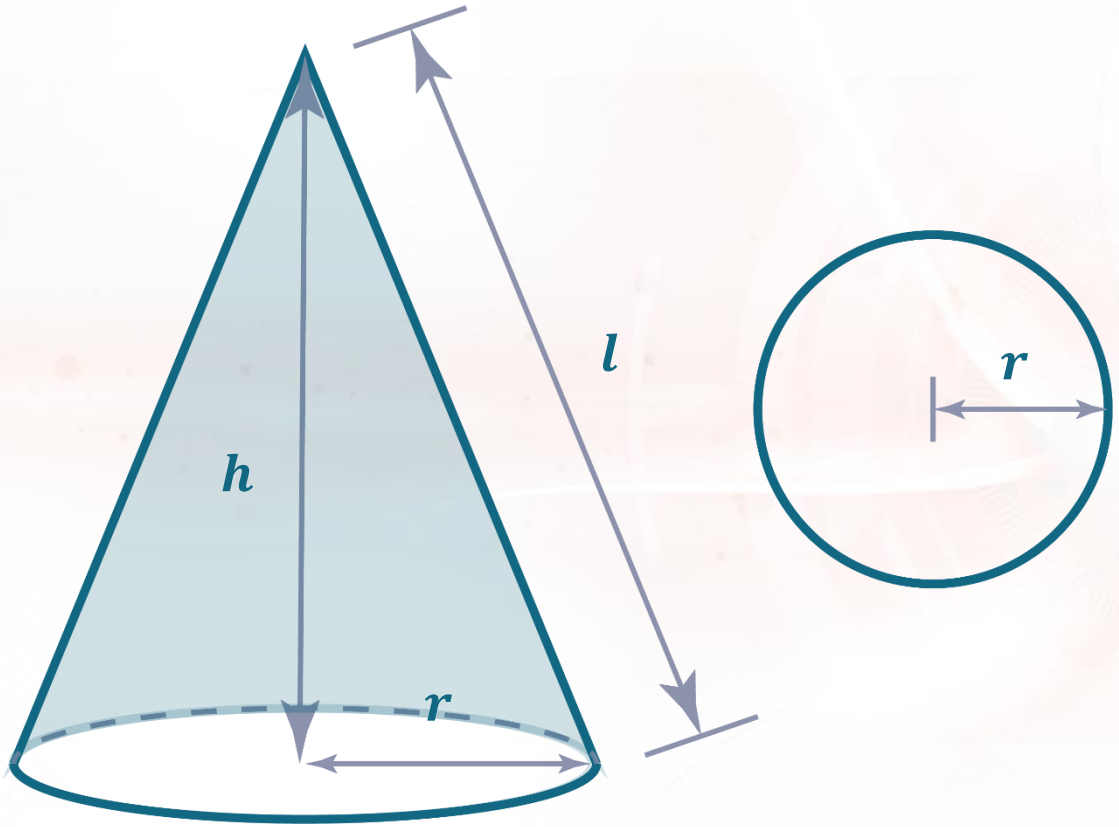
AREA OF A RIGHT CIRCULAR CONE

DEFINITION



A right circular cone is a cone whose base is a **circle** and whose axis is **perpendicular** to its base.

LATERAL SURFACE AREA



The Lateral Surface Area **S**

$$S = eP$$

$$\boxed{S = \pi r l}$$

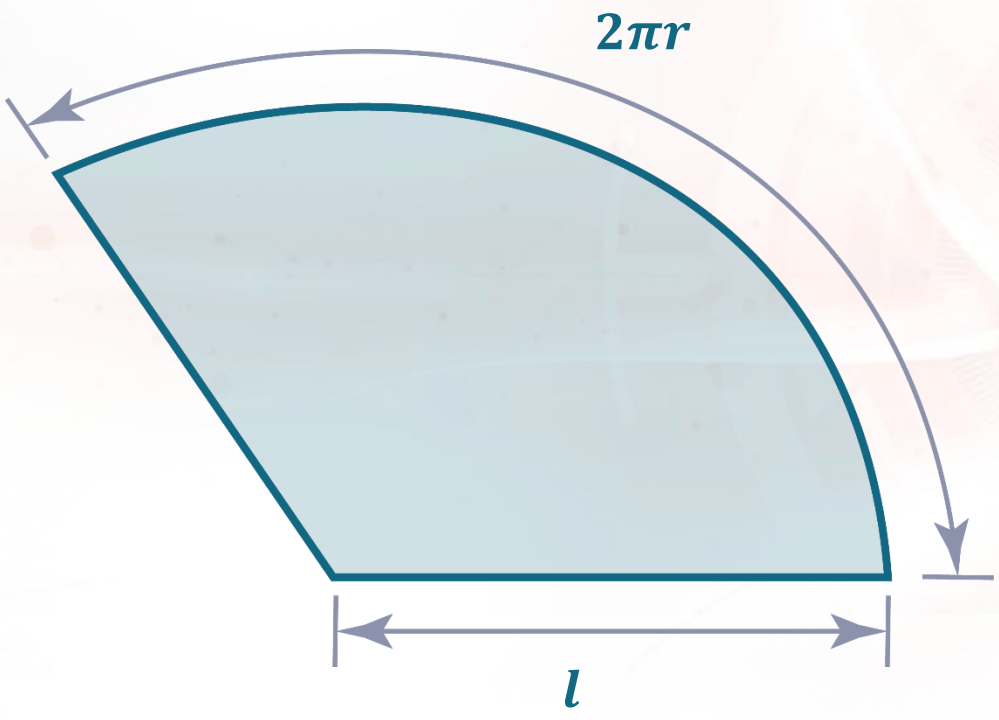
$$S = \pi r \sqrt{h^2 + r^2}$$

where

r = radius of the base

l = slant height

h = altitude



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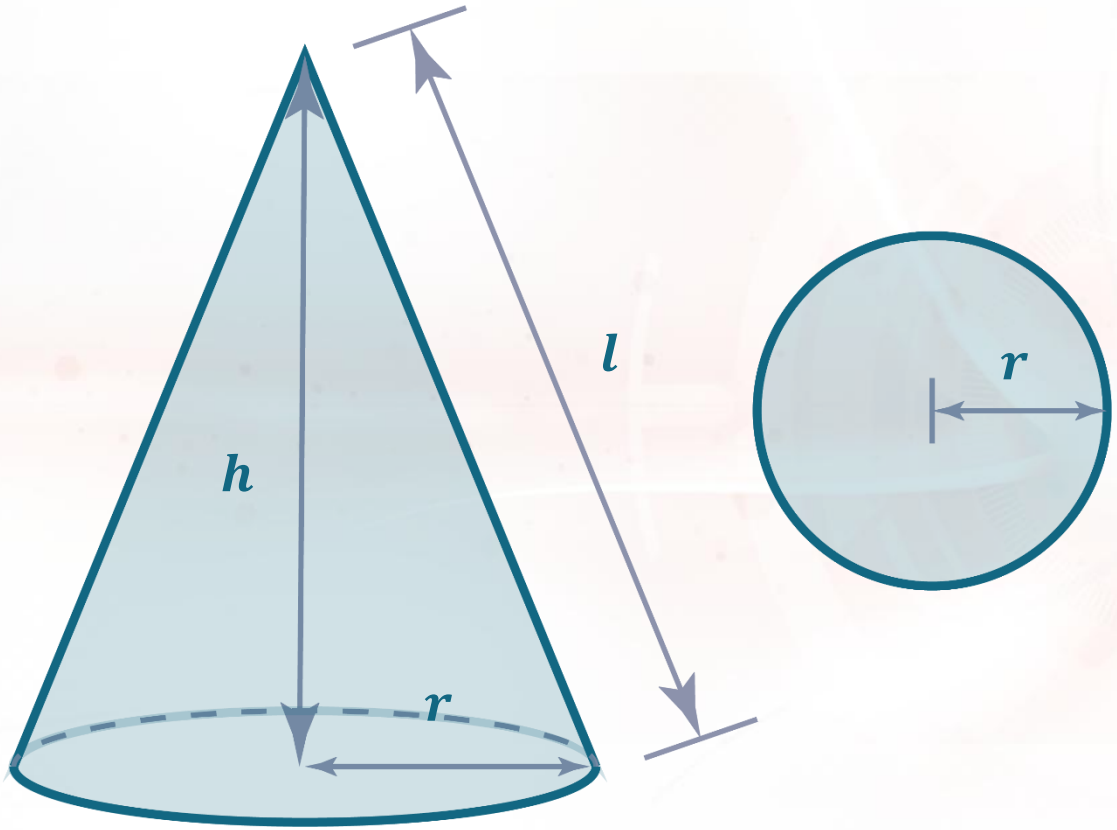
TOTAL SURFACE AREA



The Total Surface Area **S**

$$T = S + B$$

$$\boxed{T = \pi r l + \pi r^2}$$

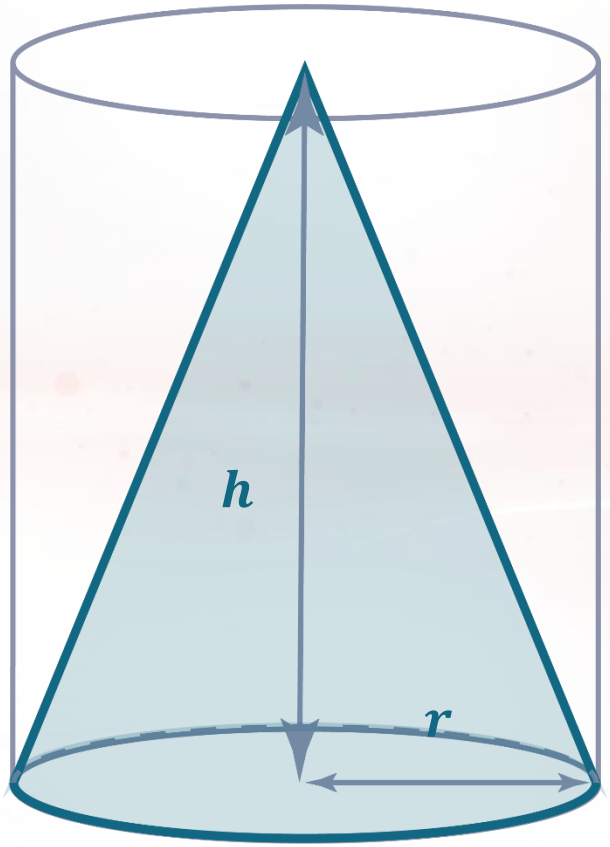
where

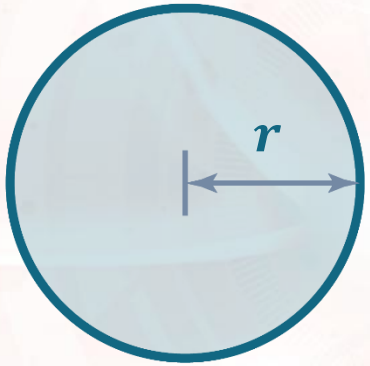
r = radius of the base

l = slant height

h = altitude

How many **square yards** of canvas will be required to make a **conical tent 15** ft. **high** and **18** ft. in **diameter**, if **10 percent** of the material is **wasted**?





The Volume **V**

$$V = \frac{1}{3}Bh$$

$$\boxed{V = \frac{1}{3}\pi r^2 h}$$

where

r = radius of the base

h = altitude

A pile of sand is in the form of a **right circular cone** of **altitude 7** ft. and **slant height 25** ft. What is the **weight** of the sand, if the sand weighs **107.5 lb. per cu. ft.**?

SEATWORK